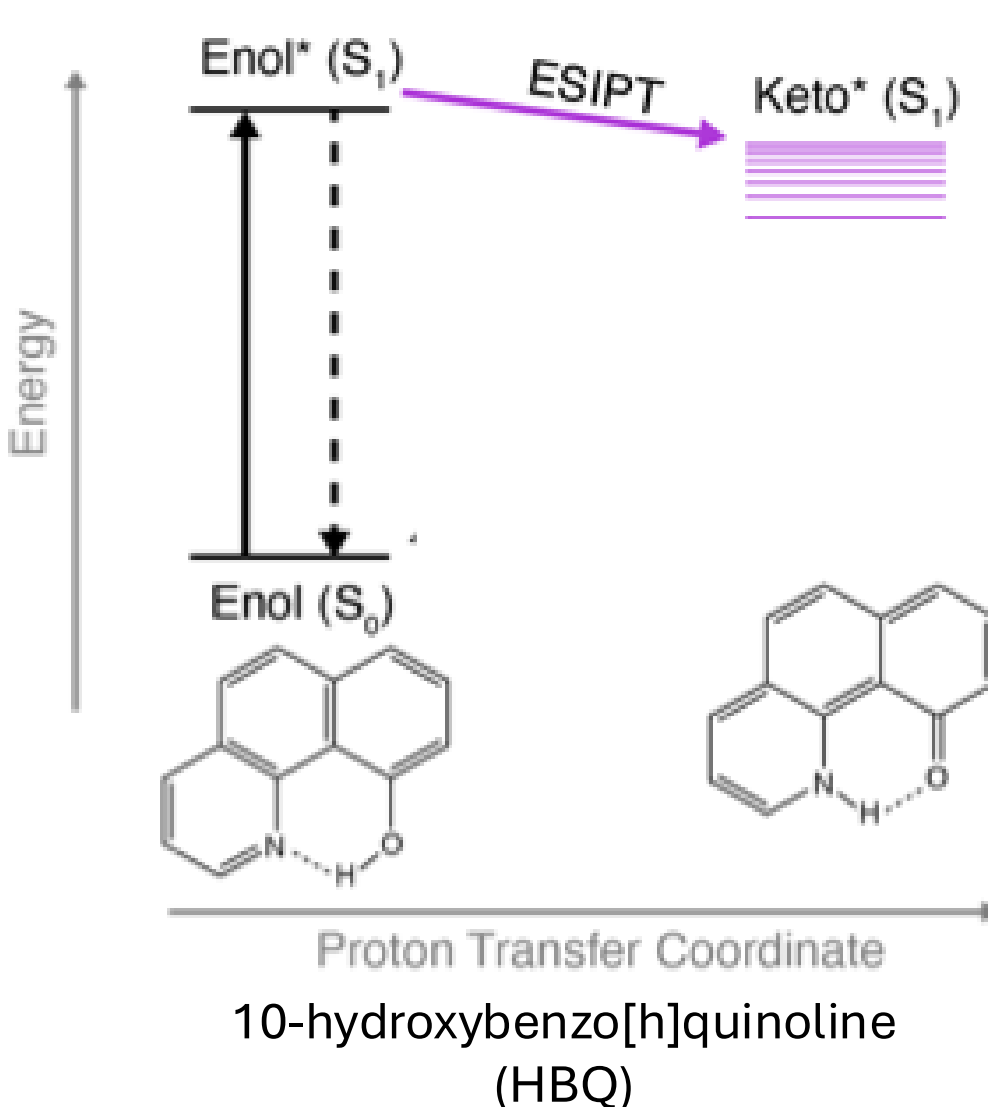


Resolving Mode Specific Vibronic Coupling from Hydrogen Bond Pre-excitation using Multi-dimensional Vibrational Electronic Spectroscopy

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Introduction and method

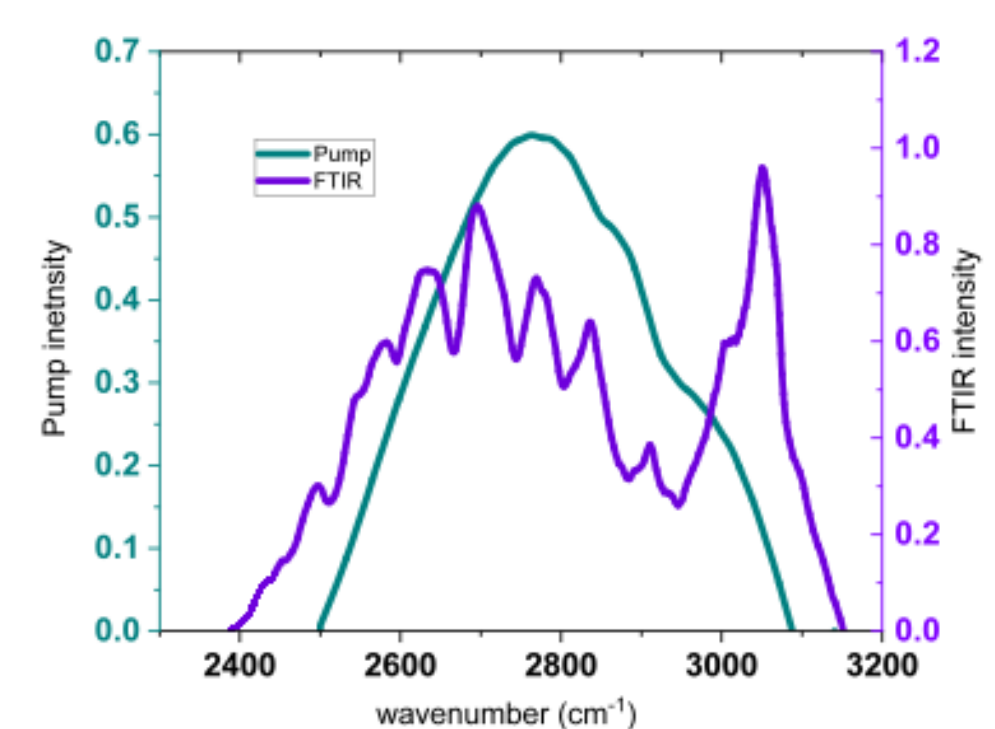


Excited state proton transfer (ESIPT) in HBQ happens extremely fast ~ 10 fs

Application : ESIPT Based Probes for Fluorescence Sensing in Materials and Biological Systems

Rational design requires molecular-level understanding of what drives the ~10 fs dynamics

Literature establishes that ESIPT in HBQ is governed by vibronic coupling. A strong interplay between electronic transitions and nuclear motion. The proton transfer is not a simple one-dimensional process; it is coupled to both high and low frequency (HF and LF) vibrational modes of the molecular scaffold.

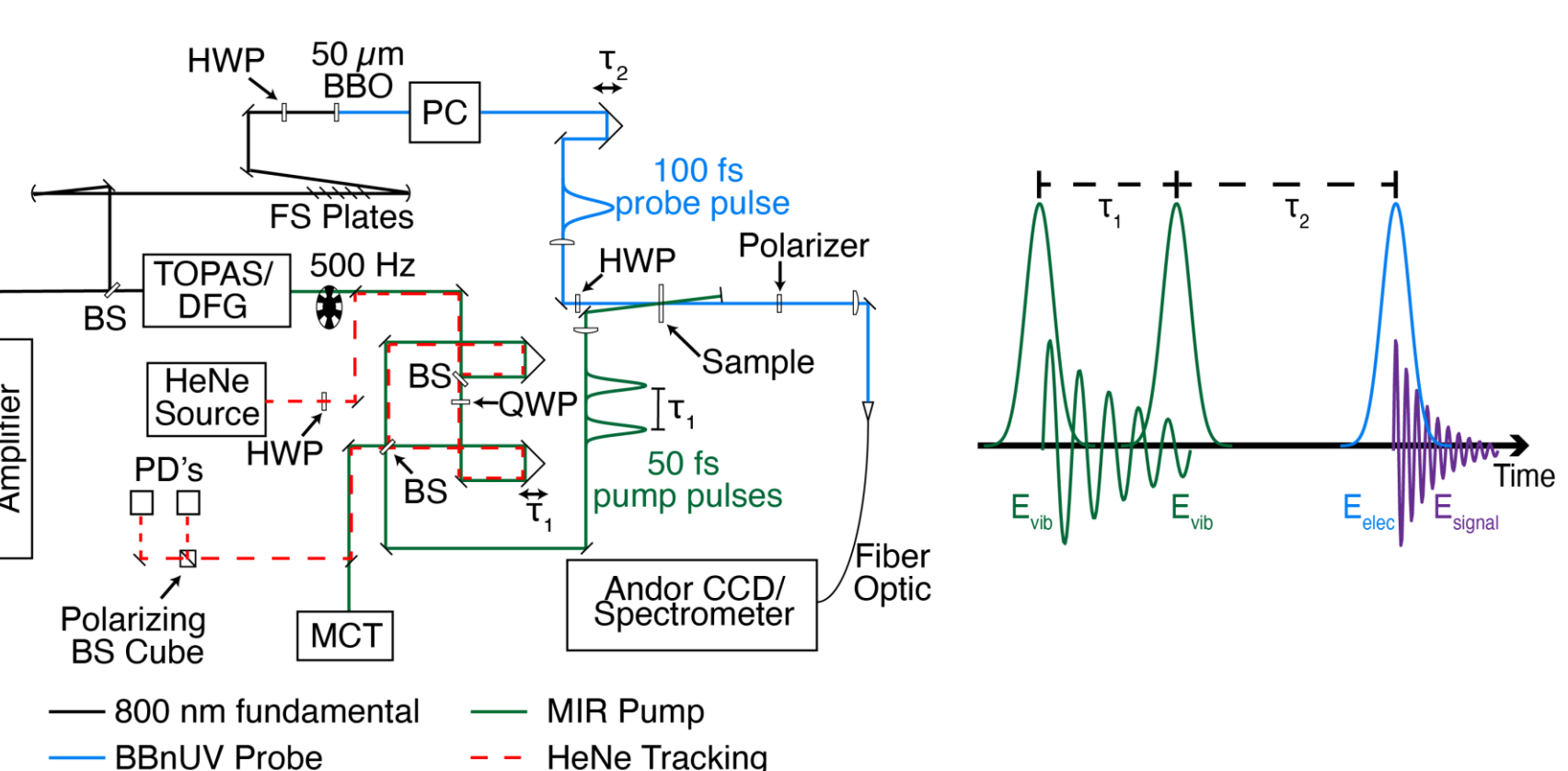


The ground-state FTIR already reveals this complexity. The O–H stretching transition is broad and highly structured, a hallmark of anharmonic coupling between the proton transfer coordinate and low-frequency molecular modes.

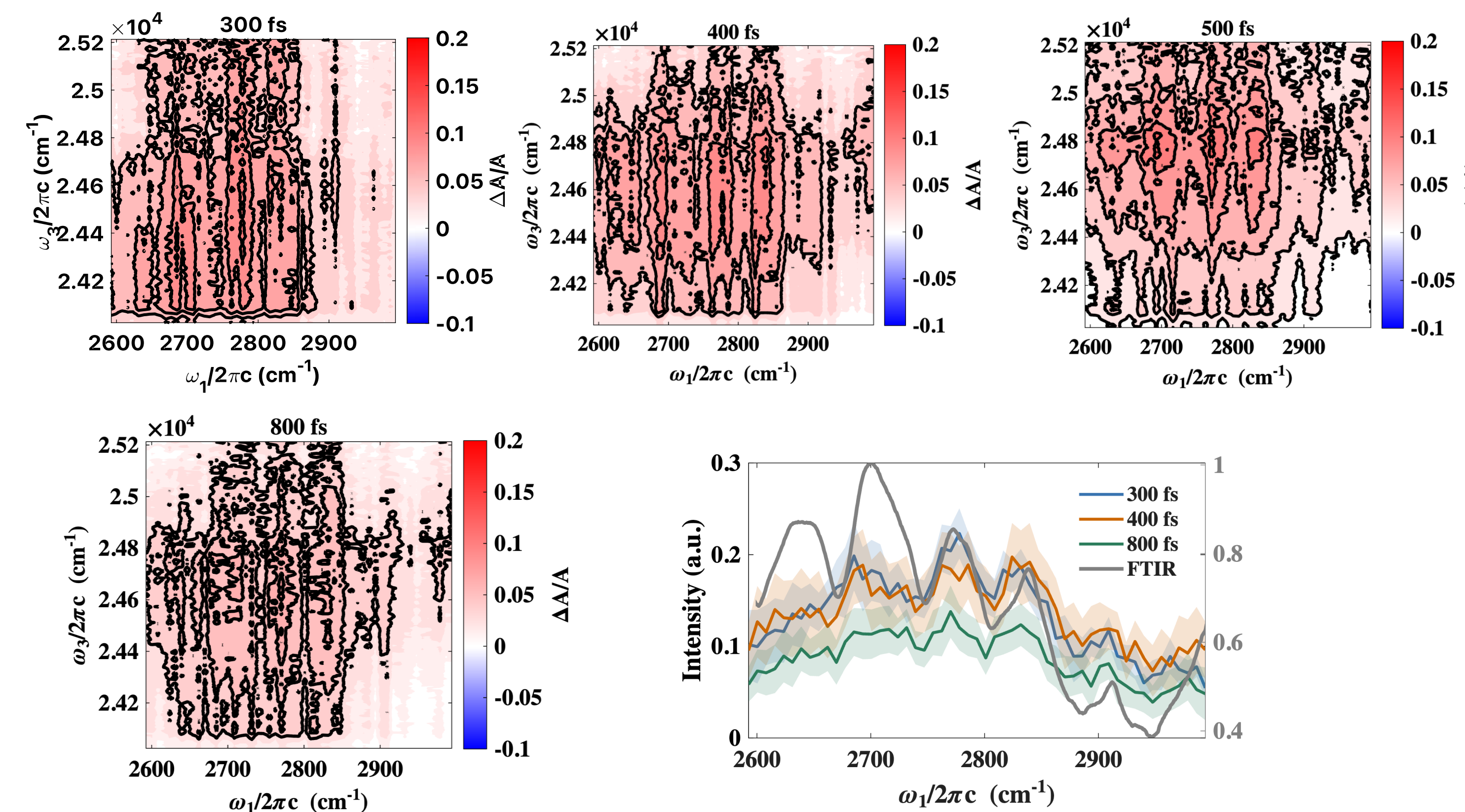
Time-dependent vibronic coupling across three disparate regimes (HF, LF, and charge transfer) remains unresolved

We propose a multidimensional vibrational-electronic spectroscopic framework to capture and control the underlying coupling mechanism

Layout

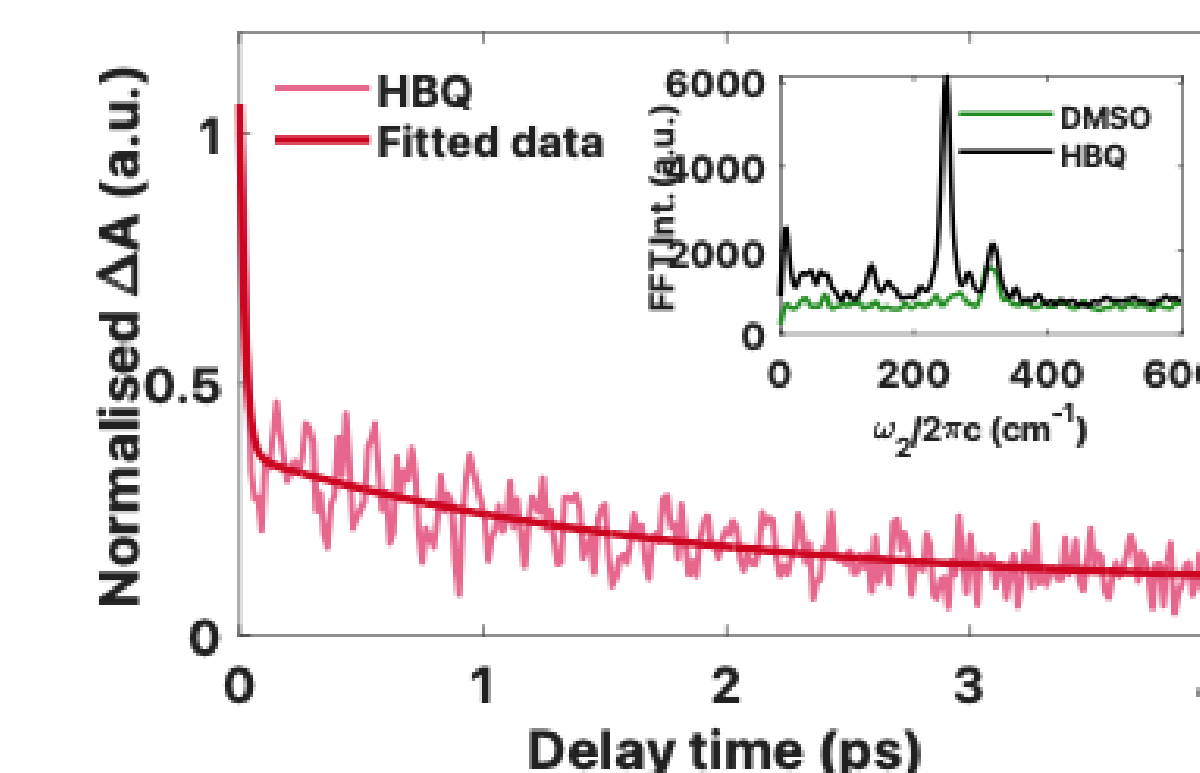


2DVE spectroscopy



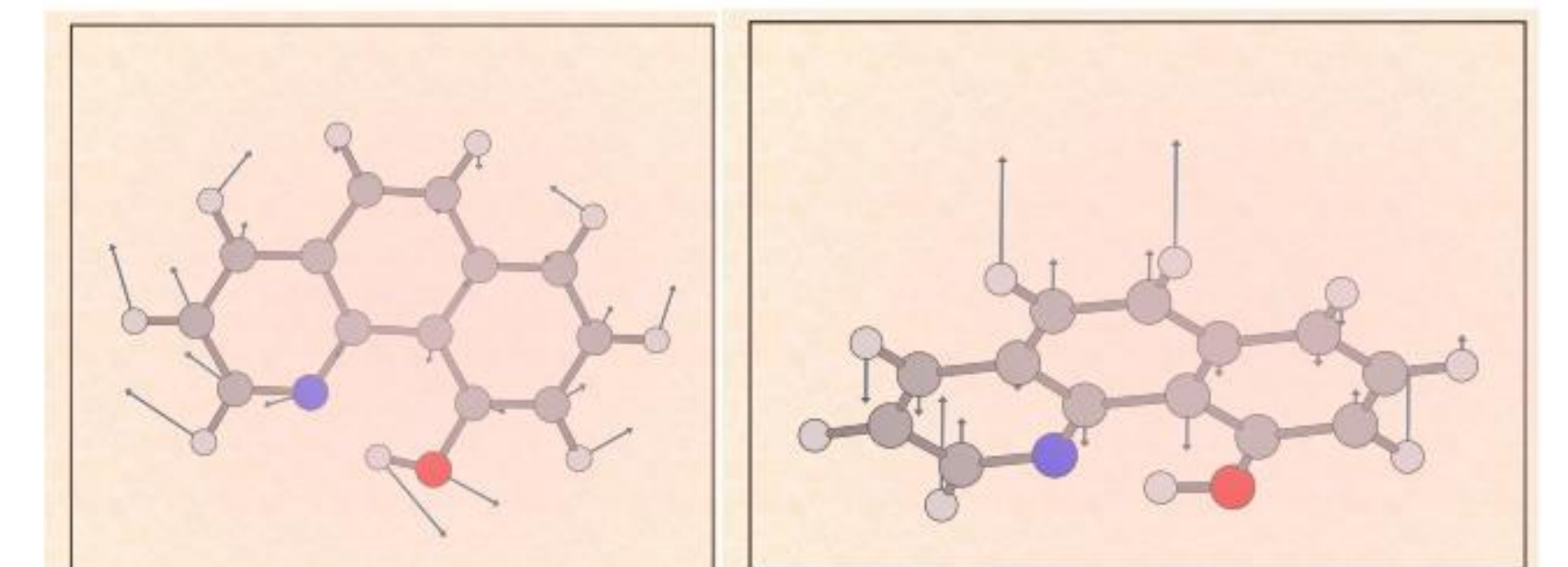
Projections along ω_1 at various τ_2 (colored lines) with SD across detection axis (shades) follow the ground-state FTIR (gray) but with oscillating relative intensities, directly reporting time-dependent vibronic coupling

Coherent oscillation shows LF coupling

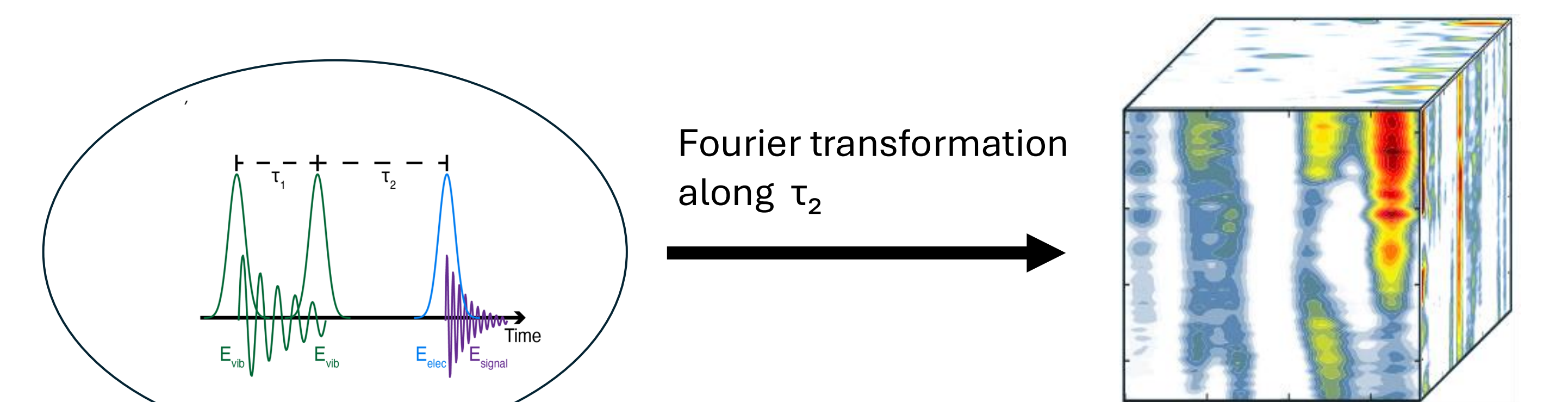


248 cm⁻¹

230 cm⁻¹



Is there a LF mode specific selective modulation ?

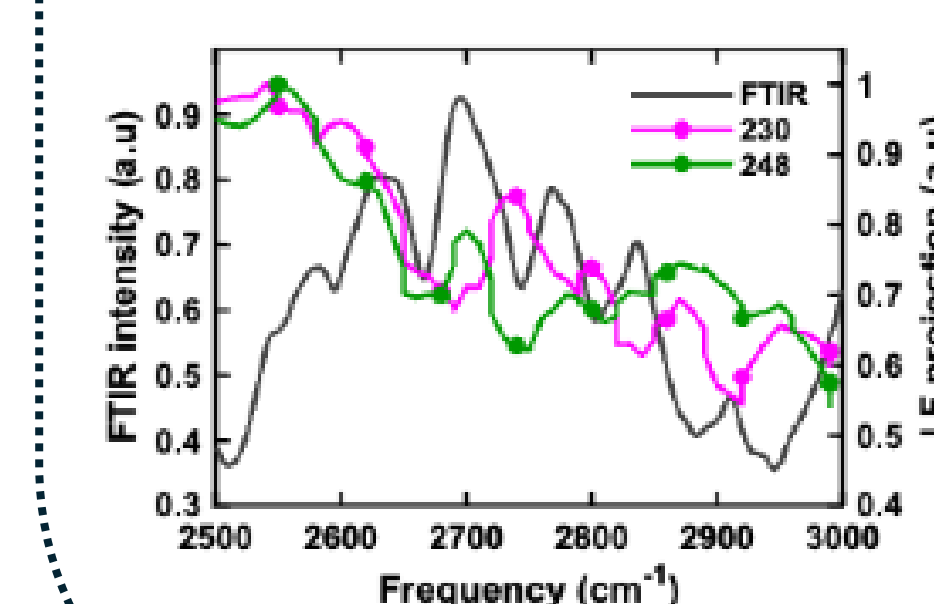
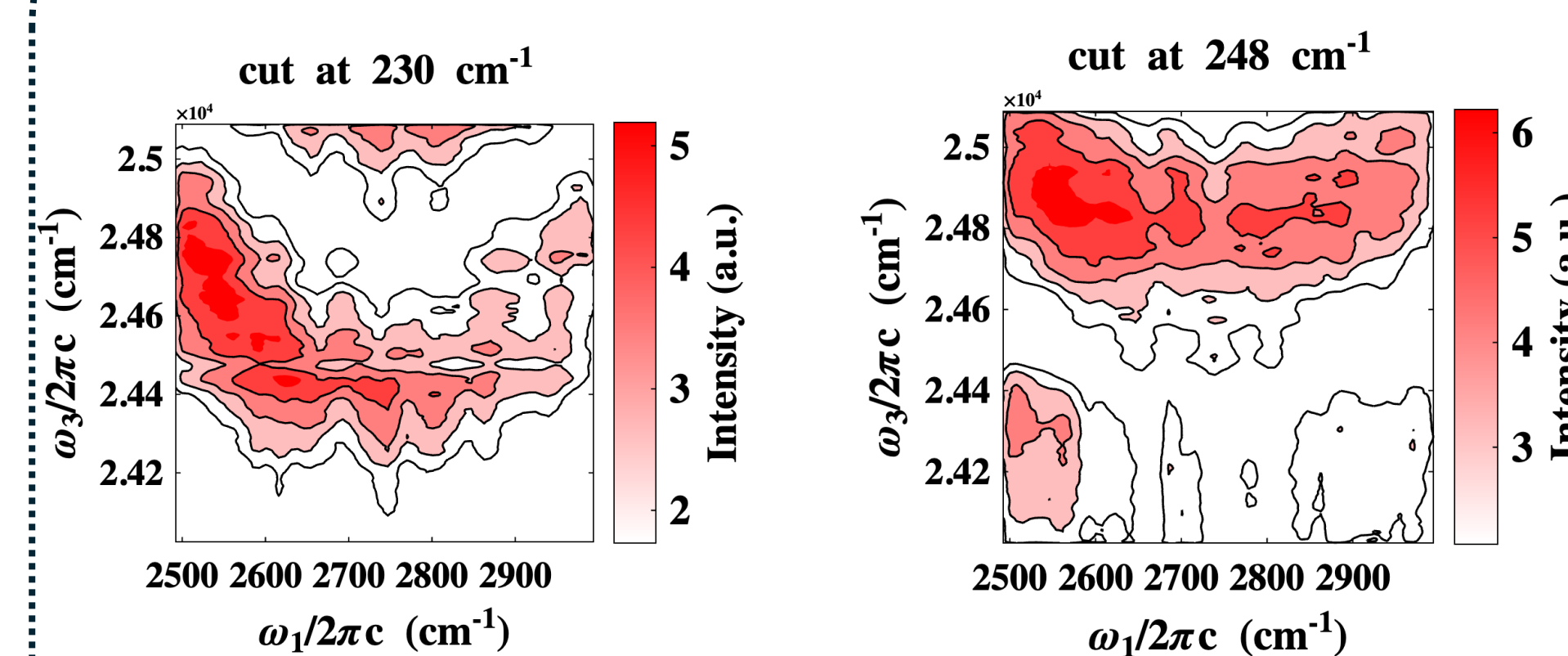


Collect multiple 2DVE data sets for a range of τ_2 value

3DVE data cube with three frequency axis (ω_1 , ω_2 , ω_3)

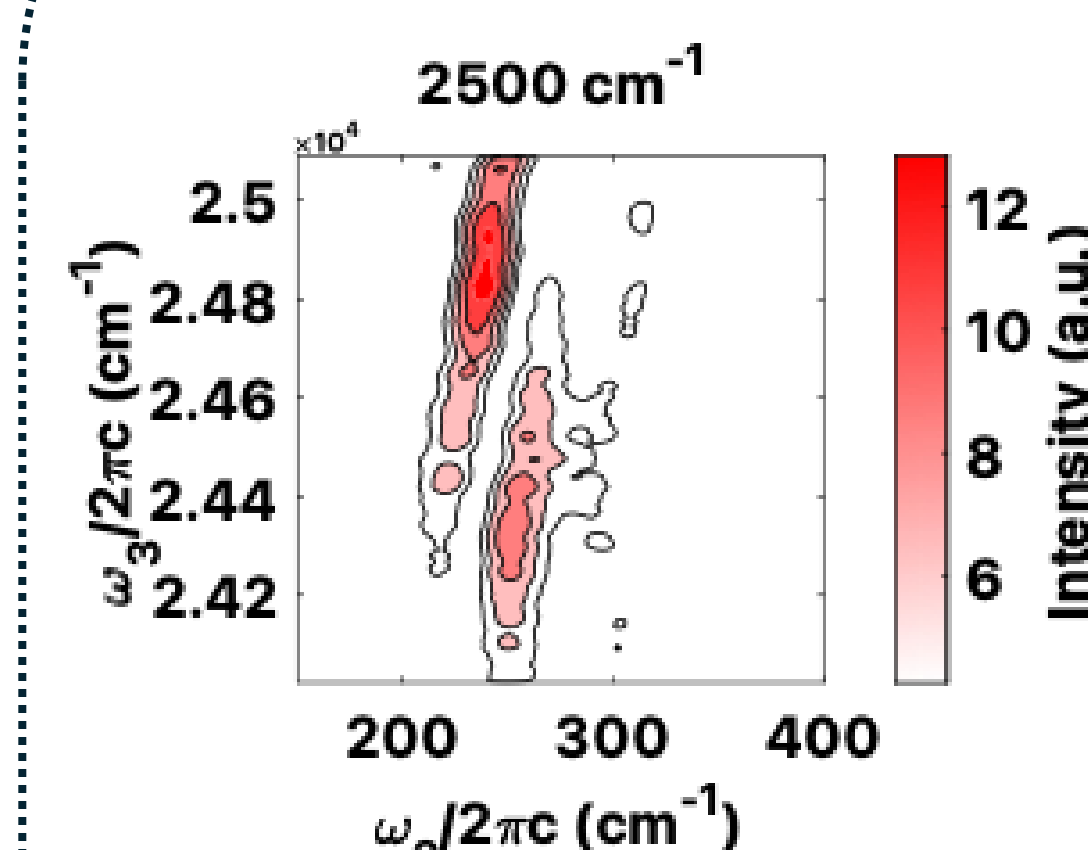
3D Fourier analysis

(ω_1 , ω_3) maps

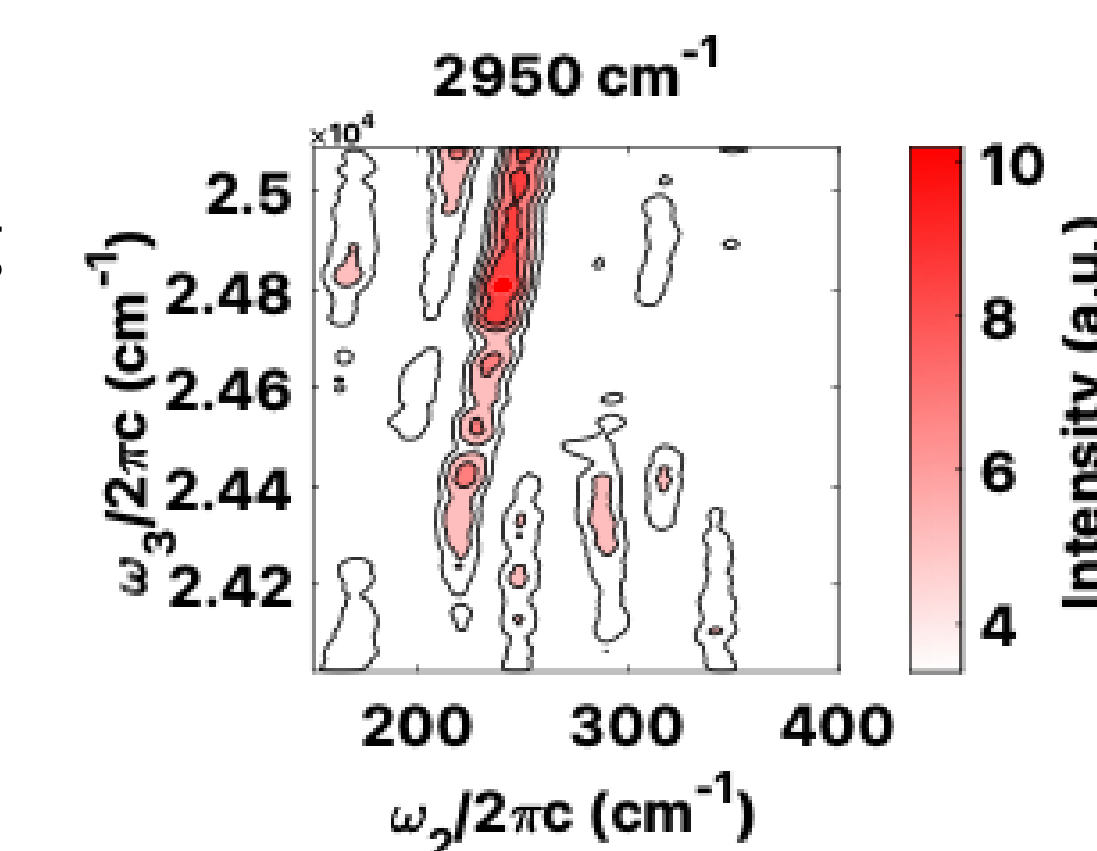


While both the projection follow the FTIR envelope, their peak maxima are anti-correlated, revealing a phase relationship arising from distinct modulation mechanisms

(ω_2 , ω_3) maps

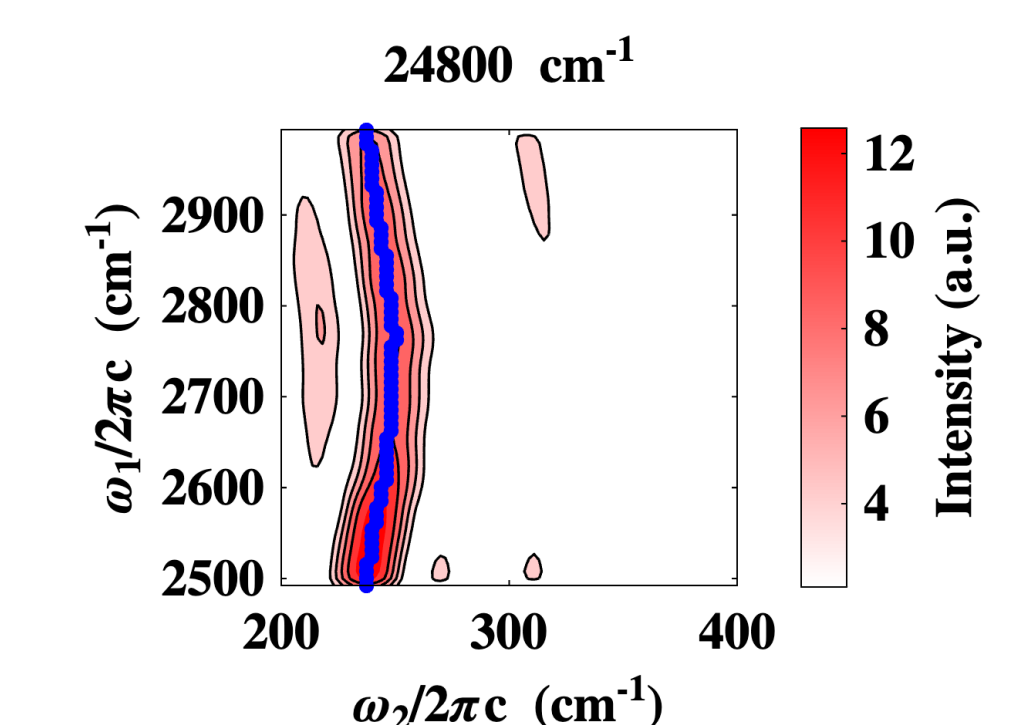


Correlation features display tilting patterns indicating coupled modulation of low-frequency modes



Anharmonic coupling responsible for mode mixing is most effective for specific combinations of O–H stretch states and LF skeletal modes

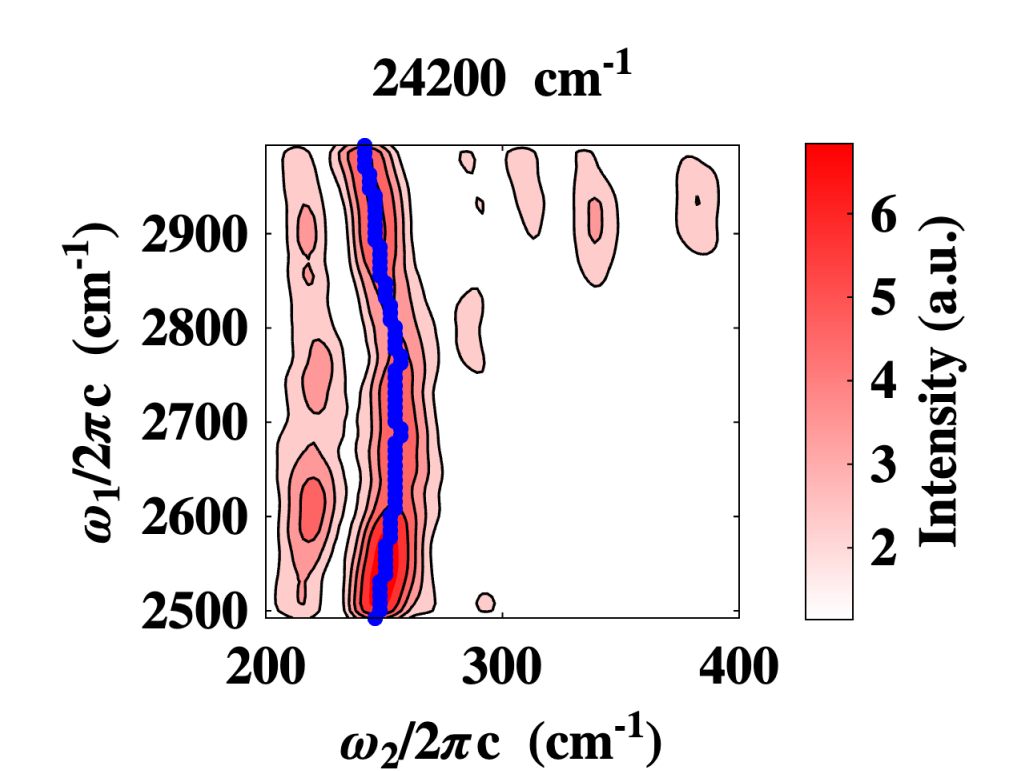
(ω_2 , ω_1) maps



Detection-frequency-dependent Fourier maps reveal nodal coupling behavior for the LF mode

Blue edge Fourier map is stronger than red edge

230 cm⁻¹ mode shows stronger modulation due to significant electron density perturbation



Conclusion

3DVE spectroscopy reveals that vibronic coupling in HBQ is mode-specific, with two discrete low-frequency modes (230 and 248 cm⁻¹) selectively coupling to the anharmonic O–H stretch along both vibrational and electronic axes, directly linking ground state vibrational structure to excited state electronic properties

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Reference

- Loe, Chatterjee, Weakly & Khalil, "Observing vibronic coupling in a strongly hydrogen bonded system with coherent multidimensional vibrational–electronic spectroscopy," J. Chem. Phys. 161, 2024
- Sedgwick et al., "Excited-state intramolecular proton-transfer (ESIPT) based fluorescence sensors and imaging agents," Chem. Soc. Rev. 47, 8842–8880, 2018

